

Battery Cost Savings Model — Client Proposal

Solar generation and battery storage for a residential household. All figures are produced by `solar_energy.ipynb` and are reproducible from `solar_energy_data.csv`.

1. Purpose

Naomi's solar panels currently serve the household only at the moment of generation; surplus is given away for no benefit. This model assesses whether installing a \$7,000 battery to store that surplus is cost effective. It quantifies the extra electricity the panels-plus-battery combination can supply, the implied dollar saving, and the net present value and internal rate of return over the battery's 20-year life under two electricity price forecasts.

2. Data and checks performed

The supplied file holds hourly generation and usage for calendar 2020. 2020 was a leap year, but no data was supplied for 29 February, so the file covers 365 days.

Completeness (all checks are assertions in the notebook; execution fails if any break)

Check	Result
Row count = 365×24 h	8,760 rows — pass
Missing values	0 — pass
Duplicate timestamps	0 — pass
Timestamps in order	monotonic increasing — pass
Supplied <code>Hour</code> column agrees with timestamp	all 8,760 rows — pass
Calendar gaps	exactly one 25-hour step (29 Feb omitted) — pass

Outlier investigation

Each correction rule is derived from the data rather than a fixed number, so the judgement can be audited.

Finding	Evidence	Action
usage = 46,000 kWh at 12:00 on 12 Jan	neighbouring hours read 0.390 and 0.570 kWh; a >20× isolated spike	replaced with mean of adjacent hours
5 negative usage readings	a household cannot consume negative electricity	replaced with mean of adjacent hours
generation = 13.05 kWh at 01:00 on 18 Apr	>5× the 99th percentile for that hour, which is 0.00 kWh because the sun is down at 01:00 on all other days of the year	replaced with mean of adjacent hours

7 of 8,760 rows were changed. Generation in hours 04:00–05:00 was examined and *retained*: it is seasonally confined to March–August and peaks in June, which is the signature of genuine dawn output, not an error. The evening readings between 20 and 61 kWh were also retained; the reasoning and the effect of removing them are quantified in section 6.

Caveat carried through to the conclusions. The corrected series implies about 49.6 kWh of daily consumption with single hours peaking at 61 kW. A household cannot draw 61 kW, and 49.6 kWh/day is roughly five times a typical residential load. The dataset is therefore not characteristic of an ordinary home, which limits confidence in the absolute dollar figures. Section 6 shows this does not change the recommendation.

3. Assumptions

Battery cost / capacity / life	\$7,000 · 12.5 kWh · 20 years (from 1 Jan 2022)	
Electricity price from 1 Jan 2022		\$0.17/kWh
Discount rate	6% p.a.	
Scenario 1	prices rise 4% p.a. (government estimate)	
Scenario 2	4.0% at 1 Jan 2023, then rising 0.25 percentage points each year (Naomi's estimate)	
Battery efficiency and self-discharge	not specified in the brief; modelled as lossless	

Degradation, feed-in tariff, replacement	not modelled
Demand and solar patterns	2020 hourly profile repeated in every year

Timing convention. Year 1 of the model is calendar 2022, priced at the 1 January 2022 tariff of \$0.17/kWh; the first escalation applies on 1 January 2023, so year n is priced at the tariff set that 1 January. Savings are treated as accruing evenly through each year and are discounted from 1 January 2022, so year n carries the factor $1/(1.06)^n$. Section 6 reports the alternative convention.

4. Method

For each of the 8,760 hours the model computes grid requirement $\max(\text{usage} - \text{generation}, 0)$, surplus $\max(\text{generation} - \text{usage}, 0)$, then dispatches the battery in the priority order the brief specifies: current generation first, then stored energy, then grid purchases. Surplus produced while the battery is already full cannot be stored and is recorded as curtailed. A single sequential pass produces every reported quantity, so no two passes can disagree.

The battery physics are solved **once**, on the 2020 data. The two price scenarios then differ only in the tariff applied to the resulting kWh of extra electricity; they are not separate physical models.

NPV discounts the annual savings at 6% and subtracts the \$7,000 outlay at $t = 0$. IRR is the discount rate at which the discounted inflows equal that outlay, solved by bisection over a wide bracket so the answer cannot depend on a starting guess, and cross-checked against an independent root finder.

5. Results

Energy

Quantity	kWh per year
Solar generated	9,768.33
used directly by the household	2,455.37
stored in the battery	3,498.61
surplus lost — battery already full	3,814.35

Bought from grid, no battery	15,640.22
Bought from grid, with battery	12,141.61
Extra electricity met by panels + battery	3,498.61
Implied saving at 1 Jan 2022 prices	\$594.76

Financial

Scenario	PV of savings NPV (net of \$7,000)		IRR	Discounted payback
1 — Government, 4% p.a.	\$9,420.98	\$2,420.98	9.42%	year 15 (2036)
2 — Naomi, 4% + 0.25pp p.a.	\$10,730.60	\$3,730.60	10.70%	year 14 (2035)

Simple payback on the undiscounted 2022 saving is 11.8 years. Both scenarios clear the 6% cost of capital, so the battery returns more than the capital costs it.

The binding constraint: the battery is larger than the surplus

The battery sits at full charge for 1,087 hours a year (12.4% of all hours), and **39% of all solar generation is discarded** because there is nowhere to put it. The specified 12.5 kWh unit captures more surplus than the household can usefully shift into the evening peak.

Capacity (kWh)	Extra kWh/yr	Generation lost	Annual saving	Simple payback
5.0	1,769.57	56.8%	\$300.83	23.3 yrs
7.5	2,436.27	49.9%	\$414.17	16.9 yrs
10.0	3,007.28	44.1%	\$511.24	13.7 yrs
12.5 (specified)	3,498.61	39.1%	\$594.76	11.8 yrs
15.0	3,906.04	34.9%	\$664.03	10.5 yrs
20.0	4,489.85	28.9%	\$763.27	9.2 yrs

Value keeps rising with capacity, so 12.5 kWh is not wasteful in the sense of buying nothing — but a client paying for more capacity should know that each extra kWh buys progressively less, and that two-fifths of the generation from an existing array is being thrown away.

6. Robustness

Three judgement calls could reasonably have gone the other way. Each was tested rather than assumed.

Alternative	NPV scenario 1	NPV scenario 2	IRR s.1	Conclusion
Reported base case	\$2,420.98	\$3,730.60	9.42%	worthwhile
All >20 kWh isolated hours treated as errors (removes 4,422 kWh of demand)	\$2,061.73	\$3,321.41	8.95%	still worthwhile
Start-of-year discounting (year n discounted n–1 years)	\$2,986.24	\$4,374.43	—	more favourable
No price escalation at all (\$0.17/kWh throughout)	–\$178.11	–\$178.11	5.68%	not worthwhile

The savings figure is driven by how much surplus solar the battery can store, not by the level of evening demand, which is why the cleaning decision moves NPV by only \$359. The one assumption the recommendation genuinely depends on is that electricity prices rise at all: held flat, the battery returns 5.68% against a 6% cost of capital and the \$7,000 is never recovered.

7. Validation

These checks execute as assertions inside the notebook, so the model cannot silently produce results if one fails.

- Completeness: row count, missing values, duplicate and unordered timestamps, agreement between the supplied hour column and the timestamps, and the single expected calendar gap.
- Corrections: no negative usage remains, and no generation reading lies outside the range physically possible for its hour of the day.
- Battery: state of charge never leaves 0–12.5 kWh; energy balance closes to 4.6×10^{-13} kWh; grid purchases never rise after adding a battery and never go negative.
- Accounting: generated = used directly + stored + curtailed, and every monthly total reconciles to its annual total.

- IRR: two independent solvers (bisection and Newton) agree to within 10^{-6} .

8. Recommendation

Install the battery, with two qualifications. Under both price forecasts the NPV is positive (+\$2,421 to +\$3,731) and the IRR of 9.42–10.70% exceeds the 6% cost of capital. The extra 3,499 kWh a year is real and repeatable.

1. **The case depends on prices rising.** At a flat \$0.17/kWh the investment returns 5.68%, below the 6% hurdle. The recommendation should be presented as conditional on the price forecasts, not as unconditional.
2. **39% of generation is currently wasted for want of storage.** If a cheaper battery can be obtained, the 10 kWh option recovers 86% of the kWh benefit for 80% of the cost, assuming price scales with capacity, and a time-of-use tariff would raise the value of every stored kWh.

Recovering \$7,000 takes 11.8 years undiscounted, or 14–15 years on a discounted basis. This is a long-recovery asset, not a quick one.

9. Suggested further work

- Confirm the 61 kW peaks with the meter. If they are genuine, this is not a residential load and the profile should be re-measured; if not, correcting them raises the modelled saving modestly (section 6).
- Obtain the battery round-trip efficiency and standby losses, currently assumed lossless.
- Investigate export or feed-in tariff rules, which would change the value of the 39% of generation currently discarded.
- Model time-of-use pricing, which would favour shifting surplus into the evening peak rather than simply storing as much as possible.

Reproduce with `jupyter nbconvert --to notebook --execute solar_energy.ipynb` using the `data_sciences` environment.